

Advanced Econometrics – Exam

June 13, 2025

Exercise 1 (30%)

This exercise is loosely based on an exercise from the textbook S. Rabe-Hesketh & A. Skrondal (2012) *Multilevel and Longitudinal Modeling Using Stata, Volume I: Continuous Responses, Third Edition, Stata Press*. The exercise uses an excerpt from a dataset from the publication Abrevaya (2006)¹

The dataset contains the following variables:

- **momid**: mother identifier (i)
- **idx**: chronological numbering of multiple children to the same mother in the database (1: first child, 2: second child, 3: third child) (t)
- **birwt**: birthweight (in grams) ($y_{i,t}$)
- **smoke**: smoking during pregnancy (1: smoking; 0: not smoking) ($x_{1,i,t}$)
- **male**: dummy variable for baby being male (1: male; 0: female) ($x_{2,i,t}$)
- **married**: dummy variable for mother being married (1: yes; 0: no) ($x_{3,i,t}$)
- **hsgrad**: DV for mother having graduated from high school (1: yes; 0: no) ($x_{4,i}$)
- **somecoll**: DV for mother having some college education but no degree ($x_{5,i}$)
- **collgrad**: DV for mother having graduated from college (1: yes; 0: no) ($x_{6,i}$)
- **black**: DV for mother being black (1: black; 0: white) ($x_{7,i}$)

Variables without the subscript t are time-invariant variables. A model of interest has the following form

$$y_{i,t} = \beta_2 x_{2,i} + \beta_3 x_{3,i} + \beta_4 x_{4,i} + \beta_5 x_{5,i,t} + \beta_6 x_{6,i,t} + \beta_7 x_{7,i,t} + \zeta_i + \varepsilon_{i,t}. \quad (1)$$

Assume $\alpha = 5\%$ significance level. Using the following outputs answer the following questions. Table 1 presents results of the Pooled OLS (POLS), random effects (RE) and fixed effects (FE) estimators.

- (5%) What is the reason that parameters for **hsgrad**, **somecoll**, **collgrad** and **black** were not estimated using the fixed effects estimator? Explain.
- (10%) What is the finding in the results of the Hausman test? Explain.
- (10%) Out of the models presented in Table 1, select the superior one. Explain.
- (5%) Let us assume that unobserved prenatal care is attributed to the individual effect. In such a scenario, should we select the random effects or fixed effects estimator? Explain. **Hint:** $Cov(\mathbb{X}\beta, \zeta)$.

```
> phtest(FE, RE)
```

```
Hausman Test
```

```
data: birwt ~ smoke + male + married + hsgrad + somecoll + collgrad + ...
```

```
chisq = 26.408, df = 2, p-value = 1.843e-06
```

```
alternative hypothesis: one model is inconsistent
```

¹ Abrevaya, J. (2006), Estimating the effect of smoking on birth outcomes using a matched panel data approach. J. Appl. Econ., 21: 489-519. <https://doi.org/10.1002/jae.851>. The JAE Data Archive directory is available at <http://qed.econ.queensu.ca/jae/datasets/abrevaya001/>

Dependent variable:			
	POLS	birwt RE	FE
	(1)	(2)	(3)
smoke	-257.334*** (16.988)	-217.024*** (18.206)	-101.214*** (29.300)
male	116.358*** (10.858)	120.944*** (9.603)	126.578*** (10.997)
married	64.952** (20.507)	71.492** (25.226)	
hsgrad	76.176*** (19.933)	87.778*** (24.642)	
somecoll	115.775*** (21.328)	127.654*** (26.344)	
collgrad	147.486*** (20.907)	160.772*** (25.756)	
black	-226.815*** (22.789)	-222.166*** (28.276)	
Constant	3,302.958*** (23.311)	3,276.547*** (27.750)	
Observations	8,604	8,604	8,604
R2	0.089	0.129	0.030
Adjusted R2	0.088	0.128	-0.804
F Statistic	119.624***	617.212***	72.591***
Note: *p<0.05; **p<0.01; ***p<0.001			

```
> pFtest(FE, POLS)
```

```
F test for individual effects
```

```
data: birwt ~ smoke + male + married + hsgrad + somecoll + collgrad + ...
```

```
F = 2.8041, df1 = 3972, df2 = 4624, p-value < 2.2e-16
```

```
alternative hypothesis: significant effects
```

```
> plmtest(POLS, type=c("bp"))
```

```
Lagrange Multiplier Test - (Breusch-Pagan)
```

```
data: birwt ~ smoke + male + married + hsgrad + somecoll + collgrad + ...
```

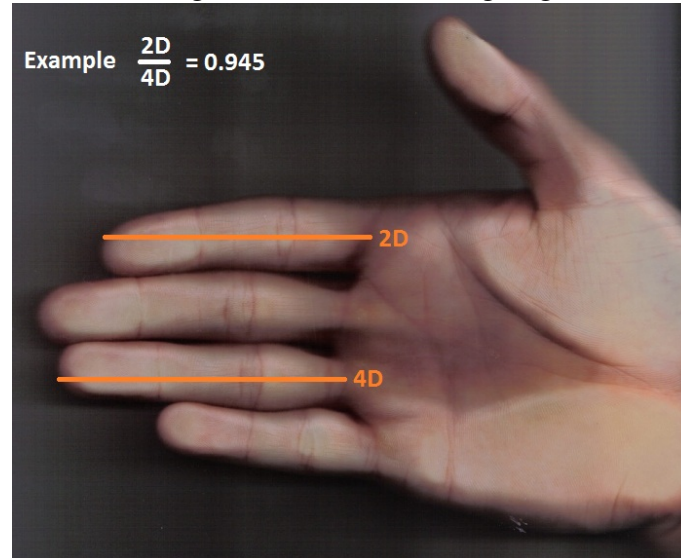
```
chisq = 1022, df = 1, p-value < 2.2e-16
```

```
alternative hypothesis: significant effects
```


Exercise 2 (30%)

Your lecturer saw a TED talk by prof. Jerzy Vetulani, top neuroscientist, pharmacologist and biochemist. In the talk, prof. Vetulani mentioned that relative length of the 2nd (index finger) and 4th (ring finger) may be an indicator of certain traits of personality.

Figure 1: Hand with index finger shorter than the ring finger. Source Wikipedia (link)



*"Research suggests that having a longer ring finger compared to index finger reflects greater exposure to male hormones during an individual's time in their mother's womb. There are differences between and within sexes in finger lengths associated with relatively more masculine versus feminine development. [...] Of course, we cannot measure prenatal hormone exposure in people. But we can use relative length of the second and fourth digits (that is, the index and ring fingers) as an indirect indication."*²

Your lecturer decided to verify the following hypothesis using a few datasets.

H_0 : persons with the ring finger longer than the index finger are less pleasant or agreeable, less likely to tolerate opposition. In other words, discussing with such a person makes no sense, and it may even have an adverse effect.

Prof. Vetulani gave an example. If you drove too fast and the police officer who stopped you has a ring finger longer than the index finger, you should not try to explain yourself or try to get away without a ticket. In such a situation, any explanation may have an adverse effect and may even result in much worse ticket.

Firstly, your lecturer decided to verify the hypothesis using a dataset of police officers. The dataset encompasses detectives, highway patrol officers, CSI investigators and foot patrol officers. The base level for the types of the officers is an CSI investigator. The variable of interest is **tektsNarrst** that is the sum of the the numbers of arrests and tickets made by an individual officer.

The independent variables were:

- **female**: gender of the officer (1: woman, 0: otherwise)
- **exper**: years of experience in being a police officer
- **highway**: dummy variable for being a highway patrol officer (1: yes, 0: no)
- **footpatrol**: dummy variable for beatcops or community officers (1: yes; 0: no)
- **detective**: dummy variable for detectives (1: detective; 0: otherwise)
- **csi**: dummy variable for crime scene investigators (1: yes; 0: no)
- **ringfnglngr**: dummy variable for having longer the ring finger than the index finger (1: yes; 0: no)

²What Can Your Hand Say About Your Cognition? Digit Length and Cognitive Abilities, <https://schaeffer.usc.edu/> (link)

The output below presents R output of the Poisson regression model.

```
> pois = glm(tcktsNarrst~female+exper+highway+footpatrol+detective+ringfnglngr,
+           family = "poisson")
> summary(pois)
```

Call:

```
glm(formula = tcktsNarrst ~ female + exper + highway + footpatrol +
    detective + ringfnglngr, family = "poisson")
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.103680	0.022620	48.792	< 2e-16 ***
female	0.008547	0.010817	0.790	0.429443
exper	-0.020557	0.001564	-13.148	< 2e-16 ***
highwayTRUE	0.094396	0.015460	6.106	1.02e-09 ***
footpatrolTRUE	0.236151	0.015026	15.716	< 2e-16 ***
detectiveTRUE	-0.062389	0.016155	-3.862	0.000113 ***
ringfnglngr	0.263514	0.011653	22.613	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 23359 on 12408 degrees of freedom
 Residual deviance: 22228 on 12402 degrees of freedom
 AIC: 53180

Number of Fisher Scoring iterations: 5

Assume $\alpha = 5\%$ significance level.

a) (6%) How to verify the hypothesis using the Poisson regression estimates? Explain.

Secondly, your lecturer used a dataset of student grades assigned by professors from a certain university. Professors with the ring finger longer than the index finger may be more assertive and less agreeable professors. This suggests that these professors are less likely to increase students' grades.

The output below presents the estimates from the ordered logit. Each observation is the grade of the individual student of a certain course. The dependent variable was **grade** and took on values {2, 3, 3.5, 4, 4.5, 5}. The list of independent variables was

- **study_time**: study time for the exam in hours
- **att_lectures**: dummy variable indicating whether the student attended lectures (1: yes, 0: no)
- **female_studnt**: dummy variable for student's gender (1: female, 0: otherwise)
- **female_prof**: dummy variable for professor's gender (1: female, 0: otherwise)
- **prof_lngr_rngfng**: dummy variable indicating whether professor has longer the ring finger than the index finger (1: yes; 0: no)
- **semester**: dummy variable for semester (1: spring; 0: winter)

Ordered Logit Regression

Log-Likelihood: -899.4653

No. Iterations: 9

McFadden's R²: 0.8168272

AIC: 1820.931

	Estimate	Std. error	t value	Pr(> t)
study_time	2.644302	0.088340	29.9332	<2e-16 ***
att_lectures	5.385563	0.215844	24.9512	<2e-16 ***
female_studnt	0.096311	0.121368	0.7935	0.4275
female_prof	0.102977	0.121588	0.8469	0.3970
prof_lngr_rngfng	-11.130835	0.387254	-28.7430	<2e-16 ***
semester	-3.134910	0.159811	-19.6164	<2e-16 ***

----- Threshold Parameters -----

	Estimate	Std. error	t value	Pr(> t)
Threshold (2->3)	42.5437	1.4390	29.565	< 2.2e-16 ***
Threshold (3->3.5)	52.9626	1.7732	29.868	< 2.2e-16 ***
Threshold (3.5->4)	61.7785	2.0620	29.961	< 2.2e-16 ***
Threshold (4->4.5)	70.7375	2.3604	29.968	< 2.2e-16 ***
Threshold (4.5->5)	80.4832	2.7043	29.761	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> lipsitz.test(ologit)
```

Lipsitz goodness of fit test for ordinal response models

data: formula: as.factor(grade) ~ study_time + att_lectures + female_studnt +

formula: female_prof + prof_lngr_rngfng + semester

LR statistic = 5.8146, df = 9, p-value = 0.7583

b) (6%) How to verify the hypothesis using the ordered logit estimates regardless of specification test results? Explain.

c) (6%) Is the conclusion from question (b) valid in the light of the specification test results? Explain.

Finally, your lecturer verified the hypothesis using credit approval data using logits and probits. The dependent variable was binary variable **approve** (1: approved, 0: not credit). The independent variables were:

- **married**: =1 if applicant married
- **unem**: unemployment rate by the industry that the applicant is employed in
- **obrat**: other obligations, percent of total income
- **loanprc**: amount of credit/price
- **male**: =1 if applicant male
- **dep**: number of dependents
- **sch**: =1 if > 12 years schooling
- **ring_longer**: dummy variable indicating whether the person responsible for credit approval had longer ring finger than index finger (1: yea, 0: no)

Dependent variable:				
	approve			
	probit1	probit2	logit1	logit2
	(1)	(2)	(3)	(4)
married	0.784*** (0.087)	0.520*** (0.097)	0.938*** (0.173)	0.964*** (0.172)
obrat		-0.028*** (0.006)	-0.053*** (0.011)	-0.054*** (0.011)
loanprc		-1.012*** (0.240)	-1.905*** (0.460)	-1.832*** (0.455)
unem		-0.037** (0.018)	-0.067** (0.033)	-0.072** (0.032)
male		-0.037 (0.110)	-0.066 (0.206)	-0.095 (0.205)
dep		-0.050 (0.039)	-0.091 (0.073)	
sch		0.015 (0.095)		
ring_longer		0.086 (0.241)	0.132 (0.446)	0.149 (0.444)
Constant	0.547*** (0.075)	2.062*** (0.316)	3.802*** (0.595)	3.706*** (0.589)
Observations	1,989	1,971	1,971	1,971
Log Likelihood	-700.877	-600.271	-600.496	-602.838
Akaike Inf. Crit.	1,405.755	1,232.542	1,232.992	1,231.676
R ²	0.053	0.187	0.186	0.183

Note: *p<0.1; **p<0.05; ***p<0.01

```
> logitgof(probit1$approve, fitted(logit), g = 10)
Hosmer and Lemeshow test (binary model)
data: credit$approve, fitted(probit1)
X-squared = 10.823, df = 8, p-value = 0.01024

> logitgof(probit2$approve, fitted(probit2), g = 10)
Hosmer and Lemeshow test (binary model)
data: credit$approve, fitted(logit)
X-squared = 12.851, df = 8, p-value = 0.1362

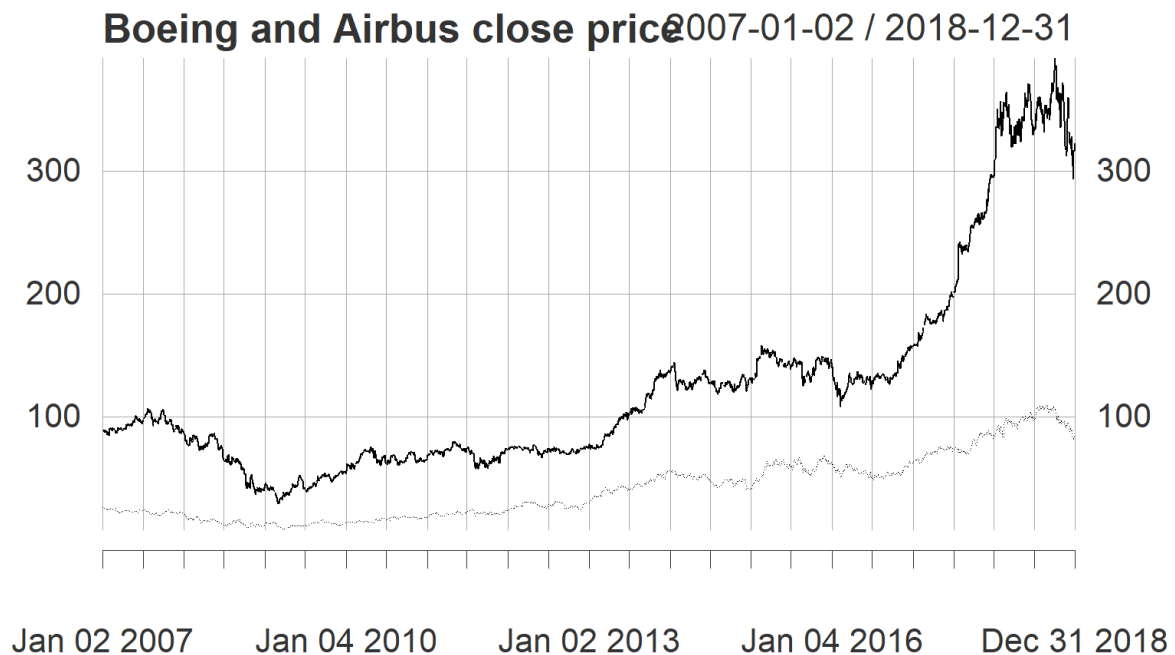
> o.r.test(logit1)
z = -0.53673 with p-value = 0.4371342
> stukel.test(logit2)
Stukel Test Stat = 4.27350 with p-value = 0.4291907
```

- d) (6%) How to verify the hypothesis using probit and logit estimates regardless of specification test results? Explain.
- e) (6%) Is the conclusion from question (d) valid in the light of the specification test results? Explain.

Exercise 3 (40%)

Your lecturer hypothesized that stock prices of the Boeing and Airbus companies may be cointegrated. The data covered daily (working days) observations for the period 2000-2029.

Figure 2: Boeing and Airbus (dotted line) close price



Use $\alpha = 5\%$ **significance level**. The critical value for cointegration is -3.67 .

- (5%) Assess which Figure out of Figure 2, Figure 3 and Figure 4 is best for visual inspection of cointegration. Explain.
- (10%) Does visual inspection prove the idea to estimate the cointegration model? Explain.
- (10%) Are these two variables cointegrated? Explain.
- (10%) Interpret the estimates of the error-correction mechanism.
- (5%) Is this relationship real or spurious? Explain.

Figure 3: Boeing and Airbus (dotted line) close price

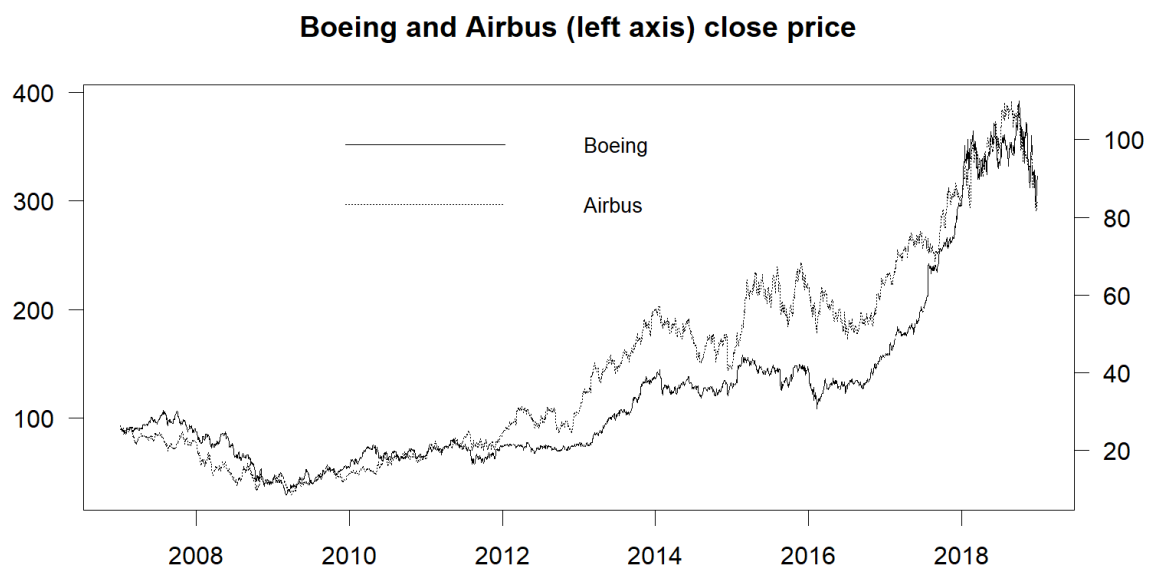
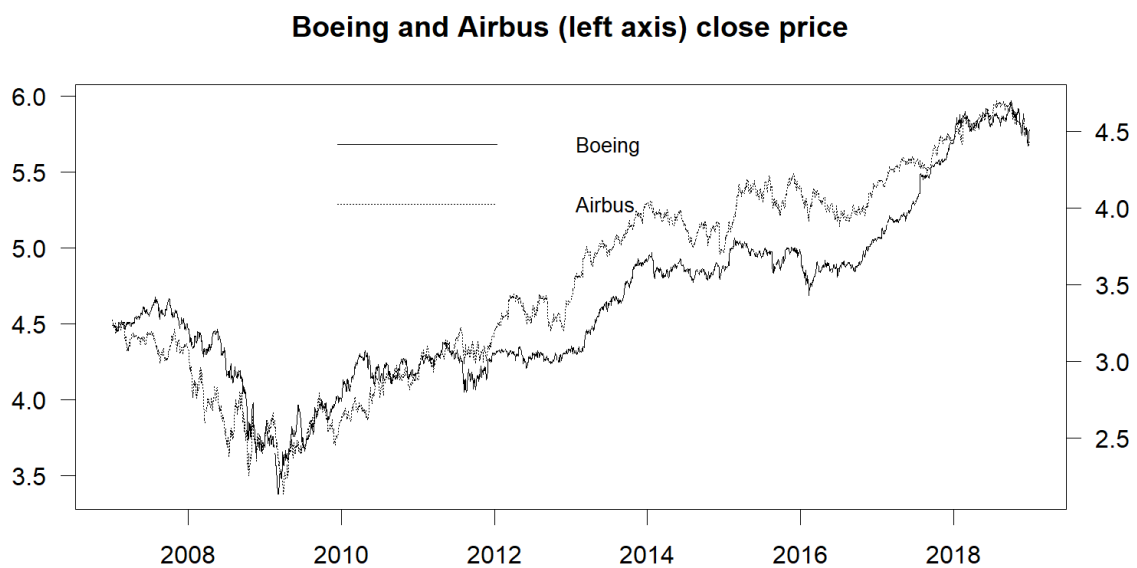


Figure 4: Logarithms of Boeing and Airbus (dotted line) close price



```
> testdf2(variable = airplanes$boeing, test.type = "c",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 0.9094522 0.99           3019      5.800892e-01
2           1 0.9433618 0.99           3018      4.276837e-04
3           2 1.0098413 0.99           3017      8.807207e-05
4           3 1.0222897 0.99           3016      2.316126e-05
5           4 0.9807238 0.99           3015      1.427279e-03
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4
1      2.8814016076      2.986900605      3.900699887
2      2.3155284326      2.399787749      3.266600026
3      0.0005099862      0.133222134      0.931484463
4      0.0002253756      0.002338404      0.796791225
5      0.0015405289      0.006880602      0.007318431
  p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.4462774      0.2367618      0.3936493      0.4196114
2      0.9835005      0.3141879      0.4936741      0.5142415
3      0.9925122      0.9997450      0.9875723      0.9200010
4      0.9961601      0.9998873      0.9999699      0.9388777
5      0.9698636      0.9992300      0.9998485      0.9999933
```

```
> testdf2(variable = diff.xts(airplanes$boeing), test.type = "nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf p_adf bgodfrey.statistic bgodfrey.statistic.1
1           0 -54.53063 0.01           2942      0.005908700
2           1 -40.29495 0.01           2941      0.001036919
3           2 -31.87489 0.01           2940      0.005877495
4           3 -26.88320 0.01           2939      0.002169279
5           4 -24.41269 0.01           2938      0.004012376
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4
1      6.941247222      7.40772260      7.97404290
2      0.001622745      0.39501330      0.95727073
3      0.006108454      0.01545360      0.61389651
4      0.004918686      0.01160760      0.01166447
5      0.011543023      0.01904559      0.02019971
  p_bg.p.value.1 p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.9387285      0.03109763      0.05997746      0.09253368
2      0.9743116      0.99918896      0.94127069      0.91620444
3      0.9388902      0.99695043      0.99949143      0.96150702
4      0.9628515      0.99754368      0.99966855      0.99998306
5      0.9494931      0.99424511      0.99930493      0.9999949
```

```
> pp.test <- ur.pp(airplanes$airbus, type = c("Z-tau"), model = c("constant"))
> summary(pp.test)
```

```
#####
# Phillips-Perron Unit Root Test #
#####
```

Value of test-statistic, type: Z-tau is: -0.1306

```
      aux. Z statistics
Z-tau-mu      0.8155
```

Critical values for Z statistics:

```
      1pct      5pct      10pct
critical values -3.43546 -2.862994 -2.567569
```

```
> kpss.test <- ur.kpss(diff.xts(airplanes$airbus), type = c("mu"))
> summary(kpss.test)
```

```
#####
# KPSS Unit Root Test #
#####
```

Test is of type: mu with 9 lags.

Value of test-statistic is: 0.2085

Critical value for a significance level of:

```
      10pct  5pct 2.5pct  1pct
critical values 0.347 0.463 0.574 0.739
```

```
> model.cooint <- lm(airplanes$boeing~airplane$airbus, data=airplanes)
> summary(model)
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      2.08810     0.96526   2.163   0.0306 *
airplanes$airbus  2.94752     0.01982 148.742   <2e-16 ***
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.17 on 2987 degrees of freedom

Multiple R-squared: 0.881, Adjusted R-squared: 0.881

F-statistic: 2.212e+04 on 1 and 2987 DF, p-value: < 2.2e-16

```

> testdf2(variable = residuals(model.coint), test.type="nc",
+         max.augmentations = 4, max.order = 4)
  augmentations      adf      p_adf bgodfrey.statistic bgodfrey.statistic.1
1             0 -2.229268 0.02506532             2988             15.3660346548
2             1 -1.995515 0.04593620             2987              0.0042559472
3             2 -1.937657 0.05187010             2986              0.0005339998
4             3 -1.868161 0.06239991             2985              0.0037139229
5             4 -1.727954 0.08364338             2984              0.0013681987
  bgodfrey.statistic.2 bgodfrey.statistic.3 bgodfrey.statistic.4 p_bg.p.value.1
1             16.082054800             17.638009417             23.72500316 0.00008856609
2              0.966062860              2.918172382              8.55764452 0.94798480755
3              0.020771855              2.135855449              7.85707579 0.98156377598
4              0.004926467              0.040928881              6.39049556 0.95140542984
5              0.003267006              0.006306862              0.01813875 0.97049364064
  p_bg.p.value.2 p_bg.p.value.3 p_bg.p.value.4
1      0.000321978      0.0005223063      0.00009067895
2      0.616910434      0.4044142102      0.07315906440
3      0.989667820      0.5446930881      0.09695624106
4      0.997539798      0.9978246092      0.17182214538
5      0.998367831      0.9998670412      0.99995912104

> lresid <- lag.xts(residuals(model))
> model.ecm <- lm(diff.xts(boeing)~diff.xts(airbus) + lresid - 1, data=airplanes)
> summary(model.ecm)

```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
diff.xts(airplanes\$airbus)	0.836367	0.046655	17.927	<2e-16 ***
lresid	-0.001094	0.001446	-0.757	0.449

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.022 on 2881 degrees of freedom

Multiple R-squared: 0.1004, Adjusted R-squared: 0.09974

F-statistic: 160.7 on 2 and 2881 DF, p-value: < 2.2e-16

